IBM Telecom Customer Churn Analysis

SQL Server | Power Bi

1. Problem Statement & Background

The telecom company analysed in this project serves over 7,000 customers across a range of contract plans, payment methods, and internet services. However, customer churn remains a significant challenge, with 26.54% of customers discontinuing service, contributing to ongoing revenue loss.

To better understand this issue, customer demographic, service, contract, billing, and churn data were analysed to identify the factors most commonly associated with customers leaving the company. The analysis focused on key areas such as contract type, tenure, payment method, internet service, and customer risk level to uncover patterns in churn behaviour.

The goal of this project is to identify high-risk customer segments, understand the business impact of churn, and provide insights that can help improve customer retention and reduce revenue loss.

Business Problem

- Why are customers leaving?
- Which customer segments are at the highest risk of churn?
- Does contract type influence customer retention?
- How does customer tenure affect churn behaviour?
- Which payment methods are associated with higher churn rates?
- What is the revenue impact of churn?
- What actions can the business take to improve retention?

2. Project Scope

The project focuses on analysing customer churn in a telecom company using customer demographic information, service usage data, contract details, and payment records.

The scope of the project includes:

- Understanding overall customer churn trends and patterns.
- Identifying customer segments at higher risk of churn.
- Analysing churn across different contract types, tenure groups, payment methods, and internet service categories.
- Evaluating the potential revenue impact resulting from customer churn.
- Developing an interactive Power BI dashboard to support business users in exploring and monitoring churn-related insights.

3. Methodology

Data Source

IBM Telecom Customer Churn Dataset

Dataset Size: 7,043 customer records

Key Data Categories:

- Customer demographics (e.g., gender, senior citizen status, dependents)
- Service information (e.g., phone, internet, and additional service subscriptions)
- Contract details (e.g., contract type and tenure)
- Payment and billing information (e.g., payment method and monthly charges)
- Customer churn status

Data wrangling

The dataset was reviewed to understand its structure and the factors related to customer churn.

Key activities included:

- Reviewing the dataset's records and columns.
- Understanding the purpose and meaning of each variable.
- Exploring customer demographics, service usage, contract details, and payment information.
- Identifying the churn status and exploring potential reasons behind customer churn.

Data Cleaning

The dataset was cleaned to ensure data quality before analysis. The following activities were performed:

- Duplicate validation to ensure each customer record was unique.
- Missing value treatment for fields such as Total Charges and Churn Reason.
- Outlier inspection to identify unusual values in monthly and total charges.
- Data consistency checks to verify categorical values and data integrity

Feature Engineering

A few additional fields were created to support the analysis and improve dashboard reporting.

Tenure Groups

- 0-1 Years
- 1–2 Years
- 2–4 Years
- 4+ Years

Charge Groups

- Low
- Medium
- High

Risk Categories

- Low Risk

- Medium Risk
- High Risk

Data Analysis

SQL queries were used to answer key business questions related to:

- Customer churn trends and patterns
- Revenue loss caused by customer churn
- Customer risk segmentation
- Customer demographics and behaviour
- Churn across contract types, tenure groups, internet services, and payment methods

Data Visualization

Power BI dashboards were created to provide:

- Executive-level KPI monitoring
- Customer churn tracking and analysis
- Customer segmentation insights
- Interactive drill-through customer analysis

4. Goals & KPIS

Goal 1: Reduce Customer Churn

KPI: Customer Churn Rate

Current Value: 26.54%

Goal 2: Improve Customer Retention

KPI: Active Customer Percentage

Current Value: 73.46%

Goal 3: Minimize Revenue Loss

KPI: Revenue Lost Due to Churn

Current Value: $139.13K

5. Technical Processes

SQL Server

- **CASE Statements:** used to create customer groups based on tenure, monthly charges, and churn risk.
- **Aggregate Functions (COUNT, SUM, AVG):** used to calculate churn rates, customer counts, revenue loss, and average charges.
- **GROUP BY:** used to compare churn across different customer segments, including contract types, payment methods, internet services, and demographics.
- **Views:** used to structure data for easier analysis.

- **Data Validation:** used to check for duplicate records, missing values, and data quality issues.
- **UPDATE Statements:** used to handle missing values and improve data consistency.
- **Filtering and Segmentation**: to identify high-risk customers and support retention analysis.

Power BI

- Developed interactive dashboards to analyse customer churn and track key business metrics.
- **Slicers and Filters:** used for dynamic data exploration.
- **Donut, bar, and line charts:** to visualize churn trends and customer segments.
- **Drill-through Analysis:** to provide detailed customer-level insights.
- **KPI Cards:** used to highlight important metrics such as churn rate, active customers, and revenue loss.

DAX

- Calculated key KPIs such as Churn Rate, Active Customers, Churned Customers, and Revenue Loss.
- Measured Customer Lifetime Value (CLTV), average charges, and average tenure.
- Compared individual customer metrics against overall averages using variance calculations.
- Generated dynamic risk insights based on contract type, payment method, tenure, and service usage.
- Supported drill-through analysis through customer-level and selected-value measures.
- Created dynamic text measures to display personalized insights and recommendations.
- Enabled interactive reporting by ensuring measures responded to slicers and filters.

6. Business Concepts Applied

- **Customer Retention:** understanding why customers leave and identifying opportunities to improve retention.
- **Customer Segmentation:** grouping customers into meaningful segments based on factors such as tenure, contract type, monthly charges, and churn risk.
- **Customer Lifetime Value (CLTV):** assessing the long-term value of customers and their contribution to overall business performance.
- **Revenue Impact Analysis:** measuring the financial impact of customer churn and the resulting revenue loss.
- **Customer Behaviour Analysis:** exploring how customer characteristics, service usage, and payment preferences relate to churn behaviour.
- **Risk-Based Customer Profiling:** Identifying customers who are more likely to churn to support targeted retention efforts.

7. Recommended Analysis & Findings

Analysis 1: Churn by Contract Type

Finding: customers on month-to-month contracts had the highest churn rate.

Impact: customers with shorter commitments were more likely to leave the company.

Recommendation: offer discounts, loyalty rewards, or bundled packages to encourage customers to switch to longer-term contracts.

Analysis 2: Churn by Tenure

Finding: customers with less than one year of tenure showed the highest churn levels.

Impact: the first year appears to be the most critical period for customer retention.

Recommendation: improve onboarding and engage new customers early through targeted support and communication.

Analysis 3: Churn by Payment Method

Finding: customers using electronic checks were more likely to churn than customers using other payment methods.

Impact: payment preferences may be linked to customer engagement and retention.

Recommendation: encourage customers to enrol in automatic payment options and offer incentives for adoption.

Analysis 4: Churn by Internet Service

Finding: Fiber optic customers experienced higher churn rates than DSL customers.

Impact: this may indicate concerns related to pricing, service quality, or customer expectations.

Recommendation: review customer feedback and service performance to identify areas for improvement.

Analysis 5: High-Risk Customers

Finding: customers with shorter tenure, month-to-month contracts, and higher churn scores were more likely to leave.

Impact: this group contributes significantly to overall customer churn.

Recommendation: focus retention efforts on high-risk customers through targeted offers and proactive engagement.

8. Project Outcome

This project helped identify the key factors related to customer churn and the customer groups most likely to leave. Using SQL and Power BI, the analysis uncovered important patterns across contract types, tenure groups, payment methods, internet services, and customer risk levels. The insights gained from this analysis can help the business improve customer retention, reduce revenue loss, and make more informed decisions. Overall, the project demonstrates how data analysis and visualization can be used to better understand customer behaviour and support business growth.